Quantum Computing 101

Unique Quantum Phenomenon

Unique properties of Quantum Computation [6] [7]

- Qubits
- Superposition
- Entanglement
- Quantum parallelism
- Quantum Tunneling

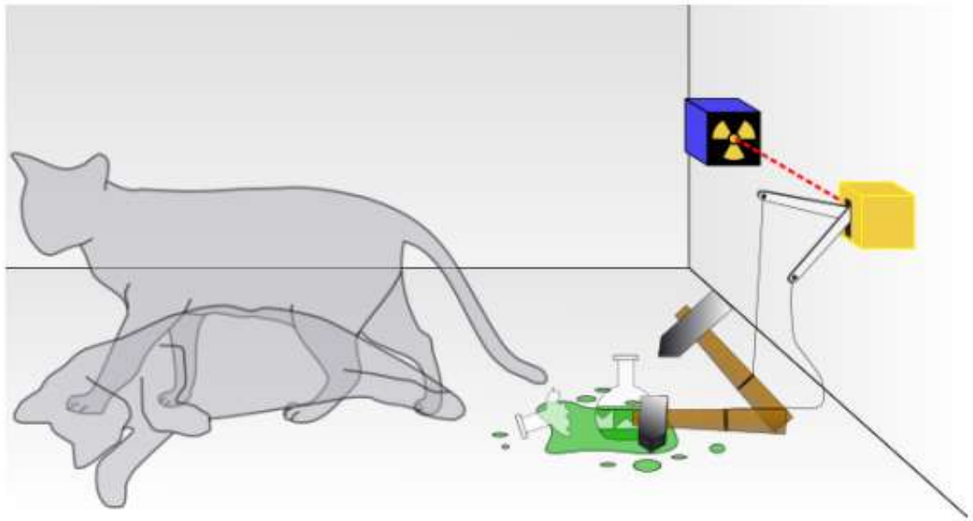
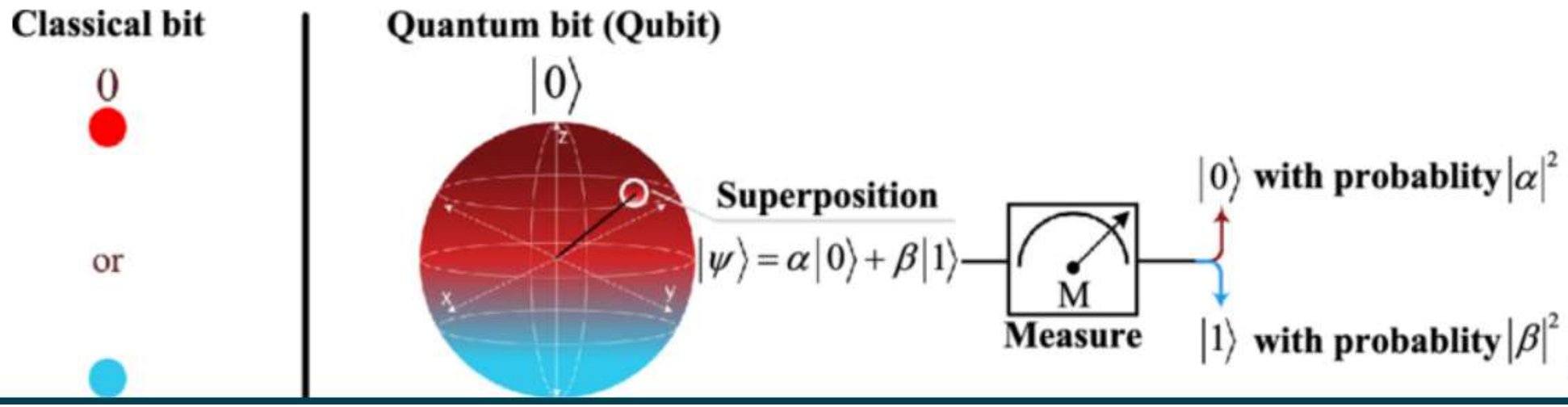


Figure 7: Schrödinger's cat



Qubit

- Basic unit of quantum information
- It is the quantum version of a bit, and its quantum state can be written and represented in the computational basis as two-dimensional vectors:-
  - $|0\rangle$ **(ket-zero)**: Represents the "0" state.

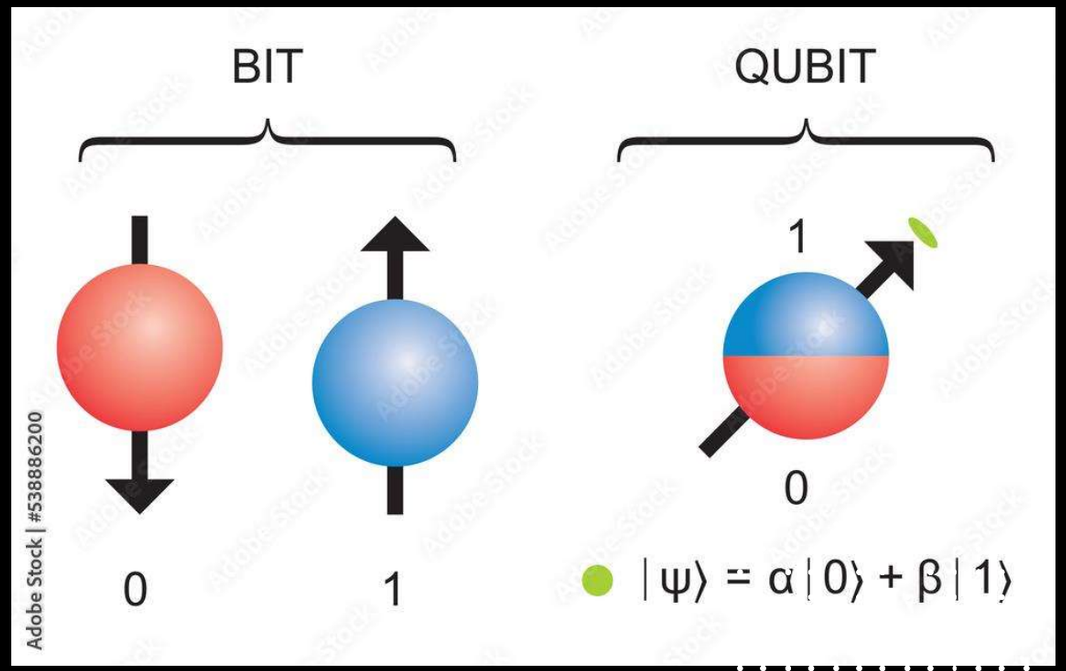
$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

  - $|1\rangle$ **(ket-one)**: Represents the "1" state.

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- These vectors form the fundamental basis states of a qubit.

- A qubit, or quantum bit, is the basic unit of information used to encode data in quantum computing.
- Qubits are typically implemented in physical systems such as superconducting circuits, trapped ions, neutral atoms, photons, semiconductor spin systems
- Quantum systems are linear systems
- The quantum states are represented by **vectors** and quantum operators are represented by **matrices**. The new quantum states are calculated by corresponding **matrix-vector multiplications.**"



Bra ket Notation

Quantum bracket notation, often referred to as Dirac notation, is a standard way to represent quantum states and operators in quantum mechanics.

- **Kets** $|\psi\rangle$ are similar to column vectors in classical linear algebra, representing the state of a system.
- **Bras** $\langle\psi|$ are analogous to row vectors, specifically the conjugate transpose of the corresponding ket. Corresponding to the ket $|\psi\rangle$, the bra $\langle\psi|$ is:
  - For $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, the corresponding bra is $\langle 0| = \begin{pmatrix} 1 & 0 \end{pmatrix}$.
  - For $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the corresponding bra is $\langle 1| = \begin{pmatrix} 0 & 1 \end{pmatrix}$.

Superposition

- Quantum system's ability to exist in multiple states simultaneously until it is measured.
- A quantum object, like an electron, can be in a superposition of different states.
- A general superposition state can be written as:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

- This should follow the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

- We have Equal Superposition State And Unequal Superposition State

Equal Superposition

- An equal superposition state means that each basis state has the same probability amplitude. For a single qubit, an example of an equal superposition state is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Here, the probabilities for each state $|0\rangle$ and $|1\rangle$ are both $|\alpha|^2 = |\beta|^2 = \frac{1}{2}$, indicating that the qubit is equally likely to be measured in either state.

Unequal Superposition

- Consider a qubit in the following state:

$$|\psi\rangle = \frac{2}{3}|0\rangle + \frac{\sqrt{5}}{3}|1\rangle$$

Here, the coefficients $\alpha = \frac{2}{3}$ and $\beta = \frac{\sqrt{5}}{3}$ indicate that the state is not in equal superposition. The probability of measuring the qubit in state $|0\rangle$ is:

$$|\alpha|^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

The probability of measuring the qubit in state $|1\rangle$ is:

$$|\beta|^2 = \left(\frac{\sqrt{5}}{3}\right)^2 = \frac{5}{9}$$

Thus, the qubit is more likely to be found in state $|1\rangle$ than in state $|0\rangle$

$$|\alpha|^2 + |\beta|^2 = \left(\frac{2}{3}\right)^2 + \left(\frac{\sqrt{5}}{3}\right)^2 = \frac{4}{9} + \frac{5}{9} = 1$$

1. $|\psi\rangle = \sqrt{1/2}\,|0\rangle + \sqrt{1/2}\,|1\rangle$
2. $|\psi\rangle = \sqrt{3/5}\,|0\rangle + \sqrt{2/5}\,|1\rangle$
3. $|\psi\rangle = \sqrt{1/3}\,|0\rangle + \sqrt{1/3}\,|1\rangle$
4. $|\psi\rangle = \sqrt{4/9}\,|0\rangle + \sqrt{5/9}\,|1\rangle$
5. $|\psi\rangle = \sqrt{7/12}\,|0\rangle + \sqrt{5/12}\,|1\rangle$
6. $|\psi\rangle = \sqrt{2/7}\,|0\rangle + \sqrt{4/7}\,|1\rangle$
7. $|\psi\rangle = \sqrt{1/10}\,|0\rangle + \sqrt{9/10}\,|1\rangle$
8. $|\psi\rangle = \sqrt{3/8}\,|0\rangle + \sqrt{1/3}\,|1\rangle$
9. $|\psi\rangle = \sqrt{9/25}\,|0\rangle + \sqrt{16/25}\,|1\rangle$
10. $|\psi\rangle = \sqrt{5/12}\,|0\rangle + \sqrt{8/12}\,|1\rangle$